ANDREW JAMES REAGAN

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Shutesbury, MA
(802) 355-3455
me@andyreagan.com
andyreagan.github.io

POSITIONS

MassMutual: Senior Data Scientist Feb 2017–current
Lead data science projects with high-impact business ROI. With Customer Journey area from start through late 2019, with projects including building state-of-the-art predictive models for customer acquisition, lead experience, advisor matching, and advisor recruiting. From late 2019 through now, working in the Risk and Product area developing and automating data science methodology for experience studies (adjusting industry mortality rates to match MM experience). These have included logistic regression, mixed effects, GAM, survival, random forest, and neural network models. Developed new standards for project structure and reproducibility. Serve as a mentor to members of the Junior Data Science program.

Smith College: Lecturer Fall 2018
Taught Statistics and Data Science (SDS) 291: Multiple Regressions to undergraduate students at Smith College in Northampton, MA. This course covers linear regression through logistic regression, with a focus on applications to real-world data science problems.

University of California Berkeley: Course design + instruction Jan 2016–current
Teach, grade, and administer two 15-seat sections of W209: Data Visualization and Communication, on a continuous basis, a core course in the Master's Program in Data Science (MIDS). Re-developed course content for Fall 2017, including recorded lectures and software tutorials for advanced D3 and user-centered visualization design.

University of Vermont: PhD Candidate Jan 2012–May 2017
Utilized the Vermont Advanced Computing Core supercomputer to perform analyses of large structered data sets (e.g. Twitter gardenhose feed). Led independent data science projects that became widely cited, rebuilt hedonometer.org, contributed to over 15 publications, and had fun doing it all. Prepared, taught, graded, and administered two 35-seat sections of calculus.

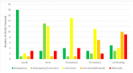
EDUCATION

University of Vermont, Burlington VT May 2017
Doctorate of Philosophy, Applied Mathematics
Advisors: Dr. Chris Danforth, Dr. Peter Dodds, Dr. Yves Dubief
Concentration: Complex Systems
Dissertation: Towards a Science of Human Stories

University of Vermont, Burlington VT Dec 2013
Master of Science, Applied Mathematics
Thesis: Predicting Flow Reversals in a Computational Fluid Dynamics Simulated Thermosyphon using Data Assimilation
GPA: 3.85

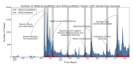
Virginia Polytechnic Institute and State University, Blacksburg VA Dec 2011
Bachelor of Science, Mathematics
GPA: 3.64, Summa Cum Laude

PUBLICATIONS

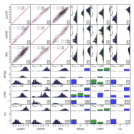


20. Social media usage patterns during natural hazards. 2019
Meredith Niles, Benjamin Emery, Andy Reagan, Peter Sheridan Dodds, Chris Danforth

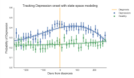
19. English Verb Regularization in Books and Tweets 2018
Tyler Gray, Andy Reagan, Peter Sheridan Dodds, Chris Danforth



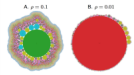
18. Divergent Discourse Between Protests and Counter-Protests: #BlackLivesMatter and #AllLivesMatter 2016
Ryan Gallagher, Andy Reagan, Chris Danforth, Peter Sheridan Dodds



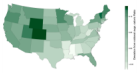
17. Sentiment analysis methods for understanding large-scale texts: A case for using continuum-scored words and word shift graphs 2017
Andy Reagan, Chris Danforth, Brian Tivnan, Jake Williams, Peter Sheridan Dodds



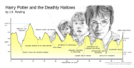
16. Forecasting the onset and course of mental illness with Twitter data 2017
Andrew G. Reece, Andy Reagan, Katharina L. M. Lix, Peter Sheridan Dodds, Chris Danforth, et. al.



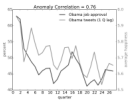
15. Simon's fundamental rich-gets-richer model entails a dominant first-mover advantage 2017
Peter Sheridan Dodds, David Dewhurst, Lewis Mitchell, Andy Reagan, Jake Williams, et. al.



14. The Lexicocalorimeter: Gauging public health through caloric input and output on social media 2017
Sharon Alajajian, Jake Williams, Andy Reagan, Stephen C. Alajajian, Morgan Frank, et. al.

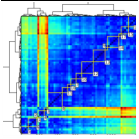


13. The emotional arcs of stories are dominated by six basic shapes 2016
Andy Reagan, Lewis Mitchell, Dilan Kiley, Chris Danforth, Peter Sheridan Dodds

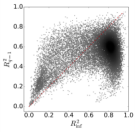


12. Public opinion polling with Twitter 2016
Emily Cody, Andy Reagan, Peter Sheridan Dodds, Chris Danforth

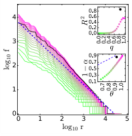
11. Tracking climate change through the spatiotemporal dynamics of the Teletherms: The statistically hottest and coldest days of the year 2016
Peter Sheridan Dodds, Lewis Mitchell, Andy Reagan, Chris Danforth



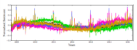
10. Game story space of professional sports: Australian Rules Football 2016
Dilan Kiley, Andy Reagan, Lewis Mitchell, Chris Danforth, Peter Sheridan Dodds



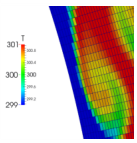
9. Selection models of language production support informed text partitioning: an intuitive and practical bag-of-phrases framework for text analysis 2016
Jake Williams, Jim Bagrow, Andy Reagan, Sharon Alajajian, Chris Danforth, et. al.



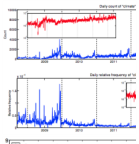
8. Zipf's law is a consequence of coherent language production 2016
Jake Williams, Jim Bagrow, Andy Reagan, Sharon Alajajian, Chris Danforth, et. al.



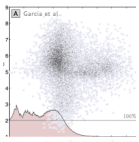
7. Benchmarking sentiment analysis methods for large-scale texts: A case for using continuum-scored words and word shift graphs 2015
Andy Reagan, Brian Tivnan, Jake Williams, Chris Danforth, Peter Sheridan Dodds



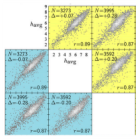
6. Predicting Flow Reversals in a Computational Fluid Dynamics Simulated Thermosyphon using Data Assimilation 2015
Andy Reagan, Yves Dubief, Peter Sheridan Dodds, Chris Danforth



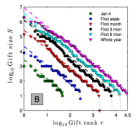
5. Climate change sentiment on Twitter: An unsolicited public opinion poll 2015
Emily Cody, Andy Reagan, Lewis Mitchell, Peter Sheridan Dodds, Chris Danforth



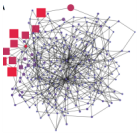
4. Reply to Garcia et al.: Common mistakes in measuring frequency dependent word characteristics 2015
Peter Sheridan Dodds, Suma Desu, Morgan Frank, Andy Reagan, Jake Williams, et. al.



3. Human language reveals a universal positivity bias 2015
Peter Sheridan Dodds, Suma Desu, Morgan Frank, Andy Reagan, Jake Williams, et. al.



2. Collective Philanthropy: Describing and Modeling the Ecology of Giving 2014
William L. Gottesman, Andy Reagan, Peter Sheridan Dodds



1. Shadow networks: Discovering hidden nodes with models of information flow 2013
Jim Bagrow, Suma Desu, Morgan Frank, Narine Manukyan, Lewis Mitchell, et. al.

SELECTED PRESENTATIONS

CIP: Developing the next generation consumer insights platform.
MassMutual DS Seminar, Amherst, MA — Fall 2019

AgentFlix: A smart matching system for advisors and leads.
Five College Stats and Data Science Meetup, Amherst, MA — Apr 2018

Julia: Time to make the jump?
MassMutual DS Seminar, Amherst MA — Apr 2018

UMass Amherst Industry Research Discussion Panel
Amherst, MA — Mar 2018

Altair: Declarative Visualization in Python, Why and How
Five College DataFest, Amherst MA — Mar 2018

Docker for DS: How it works, strengths and weaknesses
MassMutual DS Seminar, Amherst MA — Feb 2018

Bayesian banditing algorithms for intelligent pilots
MassMutual DS Seminar, Amherst MA — Sep 2017

A New Science of Stories
IBM Watson, Yorktown Hiehgts, NY — Apr 2016

Beyond Tufte: Interactive visualization for the 21st century
VT Coders Meetup, Burlington VT — Feb 2016

The emotional impact of words: comparison, critique, and benchmark of major data sets
Computer Science Research Days, UVM, Burlington VT — Sep 2015

Thermosyphon flow prediction: predicting flow reversals in a computational fluid dynamics simulated thermosyphon using data assimilation
SIAM Dynamical Systems, Snowbird UT — Jun 2015

Shapes of stories: characterizing the emotional trajectories of stories into a hierarchy of basic shapes
UVM Student Research Conference, Burlington VT — April 2015

d3.js: javascripting and data visualization
VT Coders Meetup, Winooski VT (60+ attendees) — Nov 2014

Climate Parameter Estimation with DA and Genetic Algorithms
UVM Computational Storylab Group, Burlington VT — Feb 2014

Making Beautiful Web Graphics with D3 + MATLAB
UVM Student Complexity (and Pizza) Seminar, Burlington VT — Jan 2014

Thermosyphon Flow Prediction
Master's Thesis Oral Defense, Burlington VT — Dec 2013

An Introduction to LaTeX
UVM Grad College Professional Development Seminar, Burlington VT — Jan 2013

Understanding the Natural Structure of Human Logic
Student Complexity Research and Pizza Seminar, Burlington VT — Oct 2012

Uncertainty of Rogue Wave Formation in the Nonlinear Schrödinger Equation
UVM Compuational Storylab Group, Burlington VT — Oct 2012

A Computational Approach to Iron Metabolism in Breast Epithelial Cells
Joint Mathematics Meeting, Boston MA — January 2013

Mathematically Informed Hypothesis Generation for Cancer Proliferation
Modeling and Simulation in Systems Biology REU Symposium, Blacksburg VA — August 2011

PRESS

Ups and Downs, Or how to graph a book.
Lapham's Quarterly, July 18, 2019

Teenage suicide is extremely difficult to predict. Thats why some experts are turning to machines for help.
Washington Post, September 25, 2017

Inside the Lab That's Quantifying Happiness
Outside Magazine, August 12, 2017

Researchers have quantified what makes us love Harry Potter
Washington Post, November 25, 2016

Narrative Machines
Story Grid, August 25, 2016

Story Core: UVM Researchers Use Data Mining To Identify Emotional Story Arcs
Vermont Public Radio, July 21, 2016

The Six Main Arcs in Storytelling, as Identified by an A.I.
The Atlantic, July 12, 2016

6 Story Arcs Define Western Literature, Data-Mining Study Reveals
Discover Magazine, July 6, 2016

How the New Science of Game Stories Could Change the Future of Sports
MIT Technology Review, July 27, 2015

How your social media betrays your mood
BBC, February 1, 2018

MassMutual data scientists help community make better decisions through modern analytics
MassLive, September 1, 2017

There Are 6 Basic Emotional Shapes of Stories
New York Magazine, November 28, 2016

When the stories add up: the six narrative arcs in fiction
Aeon, November 18, 2016

When computers read the canon
Boston Globe, July 22, 2016

Three, six or 36: how many basic plots are there in all stories ever written?
The Guardian, July 13, 2016

Data Mining Novels Reveals the Six Basic Emotional Arcs of Storytelling
MIT Technology Review, July 6, 2016

Twitter can tell which states love jogging & which are eating hot dogs
Washington Post, July 29, 2015

Do you live in a chocolate candy state or a green bean borough? Interactive Twitter maps reveal each states favourite snacks and activities
Daily Mail, July 22, 2015

According to the Words, the News Is Actually Good
New York Times, February 23, 2015

Spanish is the happiest language
Science Magazine, February 9, 2015

The Math of Charity
Wired Magazine, July 17, 2013

HONORS AND AWARDS

MassMutual Outstanding Mentorship Award	May 2018
SIAM Student Travel Award for Conference on Applications of Dynamical Systems	2015
University of Vermont Featured Graduate Student	2014
Virginia Tech Deans List, Every Semester attended	2008-2011
Earned Pratt Engineering Scholarship	2010
Earned Virginia Tech Scholars Scholarship	2008-2009
Earned P&C Foods Scholarship	2008
Sam Walton Community Scholar	2008
National Merit Scholar	2007
OCMTA Mathematics Contest Winner	2007

SERVICE

Mt Holyoke College guest lecturer on GAM and applications in industry	2019
Smith Hackathon and Five College DataFest session leader	2018
Pioneer Valley Planning Commission dashboard	2017
Contributions to Cookiecutter Data Science	2017
Contributions to MixedModels.jl	2017
Open source Python package for sentiment analysis (sentidict)	2017
Advisor for undergraduate students in Math 52 final projects	2015–2017
Vermont Advanced Computing Core documentation and tutorials	2015–2017
Complex Systems website database design, construction, administrator, and front end	2014–current
UVM Mathematics Department Graduate Committee	2013-2014
Student Complexity Research [and Pizza] Seminar (SCRaPS) Organizer	2013–2014
Volunteer at Bike Recycle Vermont	2013
Cycled across the USA with Bike and Build, benefiting Habitat for Humanity	2010

- My trip alone donated close to $90,000 to affordable housing and spent 10 days building with Habitat affiliates